Joseph-Vu Trinh

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Education

Virginia Tech Blacksburg, VA
Bachelor of Science - Computer Science *Graduating May 2028*

Experience

Software Developer Co-op Jan 2027 – Aug 2027
Leidos *Bethesda, MD*

Software Developer Jun 2026 – Present
Virginia Tech Department of Computer Science *Blacksburg, VA*

- Shipped backend REST API services in **Python** and **Flask** through a **CI/CD** pipeline for DBWorkout, a SQL practice platform used by **170** students in CS4604, where **90%** reported improved learning value.
- Developed the grading infrastructure that replaced manual grading, executing student queries against isolated **PostgreSQL** sandboxes via SQLAlchemy and diffing result sets against expected output.
- Integrated **OpenAI API** hint generation that turns a student's failed query and the exercise schema into a targeted hint, after students said the original feedback wasn't actionable.

Web Developer Jul 2026 – Present
SASE at Virginia Tech *Blacksburg, VA*

- Increased site visits by **53%** by rebuilding the club site in **Next.js**, **TypeScript**, and **Tailwind CSS** as a mobile-first replacement for the legacy site.
- Maintain the live site each semester, adding event listings and page updates that club leadership requests.

Undergraduate Researcher Jan 2026 – Jun 2026
Virginia Tech Department of Computer Science *Blacksburg, VA*

- Engineered a machine learning pipeline in **Python** using **OpenPose** to extract skeletal pose and gaze features from **180** one-minute classroom video segments across **300+** students.
- Improved engagement classification accuracy by **25%** over the prior approach using the extracted features, without storing identifiable imagery.
- Benchmarked LLM-generated engagement observations against human annotations using the **OpenAI API**, reducing malformed model responses by **85%**.

Full Stack Developer Intern Sep 2024 – Dec 2024
Engauge *College Park, MD*

- Built a collaborative whiteboard in **React** and **TypeScript** with **Firestore** persistence, synchronizing state across **30+** concurrent clients and contributing to a **30%** increase in student engagement during live sessions.
- Designed a **Firestore** data model giving instructors real-time visibility into every student's whiteboard in a session, with each board scoped to its student.
- Refactored the front-end component architecture of a classroom web application in **React**, cutting re-renders by about half for sessions of **150+** students.
- Implemented a live transcription and chat feature, wiring **WebSocket** updates and **GCP Cloud Functions** to persist transcripts and conversation history.

Projects

Professor Search Platform | *VTHacks* | *Python, React, Databricks Agent Bricks, GraphQL*

- Scraped and ingested **16,000** records spanning **577** professors and **3,267** reviews through asynchronous Reddit collection and paginated **GraphQL**, with filtering, deduplication, retry logic, and rate-limit handling.
- Integrated the curated dataset with **Databricks Agent Bricks** for an LLM-powered recommendation system answering natural-language questions about professors and courses, and developed the supporting React interface.

Playlist Curator | *React, TypeScript, FastAPI, Anthropic API, Spotify Web API, librosa*

- Built a playlist curation tool **end to end** in **React**, **TypeScript**, and **FastAPI** that scores Spotify tracks against user-defined themes for review before applying changes.
- Analyzed playlists of arbitrary size through **10+** backend endpoints and local audio signal analysis with librosa, combining acoustic features with Spotify metadata.

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript
Frameworks: Flask, FastAPI, React, Next.js, Node.js, Tailwind CSS
Tools & Technologies: Git, Linux/Unix, Claude Code, Codex, Figma, PostgreSQL, Docker, CI/CD, REST APIs, WebSockets, AWS, GCP, Firebase/Firestore, MongoDB (NoSQL), JUnit